

КОНСПЕКТ КНИГИ: Archaeology of Mind (Jaak Panksepp, Lucy Biven)

Формат: полный структурированный конспект по главам на основе предоставленного текста.

Идея книги: эмоции имеют глубинные эволюционные корни и опираются на древние нейронные системы мозга, общие у человека и других млекопитающих.

КЛЮЧЕВАЯ РАМКА КНИГИ

- Первичные эмоциональные системы формируются в подкорковых структурах и создают базовые аффективные состояния.
- Высшие когнитивные процессы (мышление, язык, саморефлексия) опираются на эти более древние эмоциональные уровни.
- Для понимания человеческой психики необходимо соединять данные нейронауки, эволюционной биологии, психологии и клиники.
- Исследования животных важны для объяснения человеческих чувств, мотивации, страдания и социального поведения.

Ниже – подробный конспект по каждой главе.

ГЛАВА 1. Ancestral Passions . . . certain actions, which we recognize as expressive of certain states of mind, are the direct result of the constitution of the nervous system, and have been from the first independent of the will, and, to a large extent, of habit. . . . Our present subject is very obscure, but, from its importance, must be discussed at some length; and it always is advisable to perceive clearly our ignorance.

Основной фокус:

- Анализ эволюционных и нейробиологических оснований аффективных процессов в рамках темы главы.

Ключевые тезисы из текста:

- There are many ways these ancient brain networks can make us feel—experiences we sometimes call core emotional affects and raw emotional feelings.
- So if a given brain manipulation produces emotional behavior in animals, and if human beings describe related affective experiences when they are stimulated in similar brain regions, then this complements the animal observations.
- A summary of the major differences between brain systems that mediate affective and cognitive processes in the brain.
- In sum, at present the most compelling knowledge about how emotional feelings and other affects are organized in the mammalian brain comes from direct manipulations of specific brain systems.
- Because of our higher brain expansions, we experience life at cognitive levels that other animals cannot imagine.
- Like many adaptive evolutionary developments, the brain systems that support biologically successful affective capacities have been retained as animals evolved.
- It is important to note that many of these emotional chemistries act in global ways in the brain—they are released in many brain regions to bring various network functions under the orchestration of one emotional conductor.

Практический вывод:

- Эмоциональные состояния нельзя объяснять только когнициями; они rooted в эволюционно древних мозговых механизмах и влияют на поведение, память и отношения.

ГЛАВА 2. The Evolution of Affective Consciousness Studying Emotional Feelings in

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Ключевые тезисы из текста:

- SENSORY, HOMEOSTATIC, AND EMOTIONAL AFFECTS Do animals experience primary-process affects other than the basic emotional ones?
- Overall, the assumption that emotional feelings are generated within higher cortical regions in the brain is at variance with the evidence showing that emotional systems that can elaborate rewards and punishments are located in much deeper brain regions.

- Although modern read-out theories differ from the James-Lange model in many details, the principle remains the same: The emotional states of the brain are higher brain responses to or reflections of lower brain or bodily processes.
- The affective neuroscience approach does not envision emotional feelings being “read out” by higher cognitive brain functions, although there are pervasive interactions with those regions of the BrainMind.
- They prefer to assume that emotional feelings emerge from higher regions of the neocortex (and LeDoux has claimed that he is interested in human emotional feelings as opposed to affective processes of animals).
- FURTHER SUPPORT FOR EMOTIONAL AFFECTS IN ANIMALS All animals that have been studied demonstrably like or dislike the affective feelings generated by artificial activation of the emotional systems discussed in this book.
- Its interactions with other brain regions are so extensive that it helps to facilitate most other emotional urges.

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ГЛАВА 3. The SEEKING System Brain Sources of Eager Anticipation, Desire, Euphoria, and the Quest

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Ключевые тезисы из текста:

- We have named this crucial motivational system the SEEKING-EXPECTANCY system, or the SEEKING system for short.
- PATHOLOGIES OF THE SEEKING SYSTEM A well-functioning SEEKING system is essential to physical and emotional health.
- THE CHEMISTRY OF THE SEEKING SYSTEM The SEEKING system is fueled heavily, perhaps mainly, by the neurotransmitter dopamine (DA).
- So your SEEKING system and the pigeon’s SEEKING system are aroused without the possibility of useful activity.
- When a hungry animal forages for food, its SEEKING system is aroused, but when it begins to eat, the SEEKING system becomes quiescent.
- No wonder this system is still called “the brain reward system.” In fact, this is the ancient brain system that allows us and all the other animals to gather all the rewards of the world.
- THE SEEKING SYSTEM AND A SENSE OF TIME There is, however, one very special way in which the SEEKING system is able to learn spontaneously.

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ГЛАВА 4. The Ancestral Sources of RAGE Anybody can become angry, that is easy; but to be angry with the right person, and to the right degree, and at the right time, and for the right purpose, and in the right way, that is not within everybody’s power, that is not easy.

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Ключевые тезисы из текста:

- MULTIPLE RAGE CONTROLS IN THE BRAIN, WITH MANY UNANSWERED QUESTIONS RAGE, just like every basic emotion, is regulated by many psychological processes and many brain regions.
- The Interaction of Human RAGE, Predatory Aggression, and Social Dominance It is not difficult to imagine how RAGE, predatory aggression, and social dominance might be dovetailed in the tertiary-process levels of the human psyche.
- PREDATORY AGGRESSION IS NOT DUE TO RAGE There are two major types of aggressive behavior in animals that are not pure manifestations of the RAGE system.
- THE NEUROCHEMISTRIES OF RAGE Neuroscientists know much about the brain chemicals that influence RAGE (Guerra et al., 2010; Siegel, 2005).
- Circles indicate major brain regions from which RAGE can be evoked with local

zed brain stimulation.

- At present, human investigators, even those who perform brain imaging, have not yet visualized the ways that the RAGE system and other primary-process emotional systems work.
- Although RAGE is often employed in the service of social dominance in general and inter-male aggression in particular, one should not assume that the urge for dominance is a direct expression of the RAGE system.

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ГЛАВА 5. The Ancestral Roots of FEAR Never, in his brief cave life, had he encountered anything of which to be afraid. Yet fear was in him. It had come down to him from a remote ancestry through a thousand lives. It was a heritage he had received directly . . . through all the generations of wolves that had gone before . . . Fear!—that legacy of the Wild which no animal may escape. . . . So the gray cub knew fear, though he knew not, the stuff of which fear was made.

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Ключевые тезисы из текста:

- THE INTRINSIC FEAR SYSTEM OF THE BRAIN Many people still believe that the capacity for fear is learned and that both people and animals learn to fear by anticipating danger.
- When the FEAR system is electrically stimulated in the human brain, people report fear but not pain.
- Elevated brain serotonin activity generally inhibits emotions, including fear, while less serotonin arouses emotion, including fear.
- PAIN AND THE FEAR SYSTEM Pain always arouses the FEAR system to some extent, but the reverse is not true.
- Direct stimulation simply arouses the intrinsic affective potential of the FEAR system—it arouses fear itself.
- THE BRAIN TRAJECTORY OF THE FEAR SYSTEM The existence of an intrinsic FEAR system is most directly supported by experiments that use direct electrical stimulation of the brain (for a recent overview, see Panksepp et al., 2011).
- The brain contains a number of other chemicals that activate the FEAR system.

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ГЛАВА 6. Beyond Instincts Learning and the Affective Foundations of Memory

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- Анализ эволюционных и нейробиологических оснований аффективных процессов в рамках темы главы.

Ключевые тезисы из текста:

- Once you have good solutions for evolutionary memories (e.g., the unconditional FEAR response system) as well as solid mechanisms for emotional memories (fear-conditioning), why discard them?
- In our estimation, the way in which the FEAR system mediates fear-learning remains a much under-discussed aspect of how emotional learning in the brain works.
- For sensory affects, perhaps the brain's unconditioned stimulus (UCS) processes are more important, but in considering emotional learning, we need to remember that emotional feelings are integrally anchored to the emotional action systems of the brain (FEAR UCRs).
- In a sense, the FEAR system is the conductor of fear-learning.
- Clearly, emotional arousal is a necessary condition for the creation of FEAR-learning memories—for example, freezing responses to previously neutral stimuli—because animals can only be conditioned if the training procedure evokes affective arousal.
- Higher cognitive experiences also play no essential role in the success of brain-conditioning mechanisms that result in emotional learning and memory.
- THE ACTUAL MECHANISMS OF MEMORIES Most of learning-memory research has proceeded

ed without much concern about emotional issues.

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ГЛАВА 7. LUSTful Passions of the Mind From Reproductive Urges to Romantic Love

Основной фокус:

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Ключевые тезисы из текста:

- We have seen that sexual circuitry and sexual chemistry in male and female brains are different.

- GENDER DIFFERENCES WITHIN THE BRAIN/MIND Male and female brain systems are somewhat different in terms of their sexual urges, satisfactions, and many other psychological traits.

- A female brain in a male body will result in a person who feels like a woman in the body of a male, and a male brain in a female body produces a person who feels like a man in the body of a woman.

- There are abundant subcortical brain sites that generate both sexual arousals and brain stimulation reward (Caggiula, 1970; MacLean & Ploog, 1962), highlighting the widespread locations of sexual pleasure.

- MALE AND FEMALE MINDS: FETAL DEVELOPMENT OF GENDER WITHIN THE BRAIN Developmental processes during the second trimester of gestation seem crucial in establishing the sexual identity of the human brain (Murray et al., 2000).

- Sexuality is further complicated (especially at the tertiary-process level for humans) because the sexual body and the sexual brain develop along different trajectories in utero.

- Testosterone and Male Aggression In addition to facilitating male sexual responses, testosterone also plays a crucial role in generating the male's aggressive urges for social dominance.

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ГЛАВА 8. Nurturing Love The CARE System Tiny eyes that rode inside me,

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Ключевые тезисы из текста:

- Clearly, the maternal CARE system helps the brain to rapidly learn much about maternal competence.

- For example, while oxytocin clearly contributes to the initiation of maternal urges, once maternal behaviors have been established, oxytocin is no longer critically important for competent maternal behavior.

- Furthermore, although oxytocin is not essential for producing maternal behaviors in experienced mothers, maternal behavior is weak if oxytocin systems are generally unresponsive (Nishimori, et al., 2008).

- There are abundant reasons to believe that many maternal chemicals play central roles in the maternal urges of human mothers.

- When mother rats have had past maternal experience, oxytocin is not essential in the short run for competent maternal behavior.

- Under normal conditions, even after maternal behaviors have been learned, oxytocin is still released in the mother's brain.

- Thus, chemicals other than oxytocin must play essential roles in the creation of maternal impulses.

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ГЛАВА 9. Born to Cry The PANIC/GRIEF System and the Genesis of Life-Sustaining Social Bonds

Основной фокус:

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Ключевые тезисы из текста:

- Brain opioids, in addition to oxytocin, and probably brain prolactin as well, are secreted onto the chemical receptors of needy GRIEF networks when people and animals touch each other and form positive social bonds.

- Positive social interactions release brain opioids, providing positive social feelings, and placebo effects may reflect, in part, the perception that mental health professionals and other significant others are caring about one's depressive feelings.

- A little separation anxiety may help solidify social memories and social bonds

- In the brain of an infant rat, when social bonding is crucial to survival, one finds high levels of oxytocin receptors in the brain structures of the GRIEF system.

- Brain imaging of sadness in human adults has highlighted very similar brain regions, accompanied by a reduction in brain opioid activity (Zubieta et al., 2003).

- It is even possible that much of the emotional attraction and power of religion is based upon social attachment systems, no doubt supplemented by other social feelings of the brain (Thandeka, 2005, 2009).

- Perhaps such children develop unresponsive brain oxytocin or opioid systems.

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ГЛАВА 10. PLAYful Dreamlike Circuits of the Brain The Ancestral Sources of

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Ключевые тезисы из текста:

- Although psychostimulants decrease play, dopamine systems are nevertheless aroused during normal PLAY (Panksepp, 1993).

- Just as the urge to PLAY builds up systematically during social isolation or other periods of play deprivation, the desire for play systematically diminishes when pairs of juvenile lab rats are allowed free play for half an hour (Burgdorf et al., 2006; Panksepp & Beatty, 1980).

- Second, many misinterpret PLAY as a form of aggression, as reflected in the common label "play-fighting" for rough-and-tumble play (Aldis, 1975).

- Compared to animals that have had little history of play, those with a history of abundant play experiences spend more time with others suggesting a social-bonding (friendship) function of play.

- Surely, play increases reproductive fitness in various ways, but sexual-type behaviors are very infrequent during the course of rough-and-tumble play in rats, even though various other animals show quite a bit more mounting behavior during play.

- Animals whose play-instigation areas are anesthetized simply can't coordinate their play urges any more.

- Other brain lesions arouse emotional states that inhibit play.

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ГЛАВА 11. Toward a Neurobiology of the Soul The Core SELF and the Genesis

Основной фокус:

- Анализ эволюционных и нейробиологических оснований аффективных процессов в рамках темы главы.

Ключевые тезисы из текста:

- Central to our hypothesis is the assertion that midline systems in the brain, which give all mammals a universal (nomothetic) core SELF, can support various renditions of the self (idiographic forms) in other regions of the brain related to higher information processing.

- The core SELF nevertheless seems clearly related to primary-process emotional and other affective processes of the BrainMind.
- SELF-RELATED PROCESSING IS GROUNDED IN LOWER BRAIN MOTOR FUNCTIONS What are the neuroevolutionary foundations of the core SELF?
- THE CORE SELF AND THE MECHANISMS OF AFFECTIVE CONSCIOUSNESS The kind of consciousness experienced by the core SELF is fundamentally affective, without any propositional content.
- Therefore, an affective core SELF may provide a solid neuroevolutionary scaffolding for all higher self-functions.
- Thus, the biophysical similarity between affective network-firing dynamics in the core SELF and instinctual emotional actions allows us to study subjective emotional feeling in animals through a study of their objective, instinctual emotional expressions.
- The core SELF is massively connected to many other parts of the brain—to various sensory triggers and regulatory feedbacks, to motor functions, autonomic integrative responses, as well as to many higher cognitive processes, especially of cortical midline systems.

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ГЛАВА 12. Brain Emotional Systems and Affective Qualities of Mental Life

Основной фокус:

- Анализ эволюционных и нейробиологических оснований аффективных процессов в рамках темы главы.

Ключевые тезисы из текста:

- EMOTIONAL DYNAMICS AND AFFECTIVE BALANCE THERAPIES Psychotherapy can benefit from our increasing understanding of the emotional-instinctual action dynamics of the brain.
- Unconditioned emotional dynamics provide a scientific way to understand how primary-process emotional feelings are actually generated in the brain.
- When the subcortical emotional powers of the human brain become tempestuous (or dysregulated beyond understanding), overwhelming and often lasting emotional problems can emerge.
- Primal emotional feelings go hand in hand with emotional action dynamics, constituting distinct varieties of mental experience.
- In any event, practitioners should clearly envision the natural primal emotional systems of the mammalian brain, and they should conceptualize human emotional problems, at least partly, in those terms.
- Depression may arise when certain primary process emotional systems become chronically imbalanced.
- At present, analyses of mammalian brain emotional systems, with attempts to model psychiatric disorders in light of affective circuit imbalances, remain rare (Panksepp, 2010b).

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ГЛАВА 13. Coda Philosophical Reflections: Can We Go From Mice to Men

Основной фокус:

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Ключевые тезисы из текста:

- The brain's emotional foundations of human personality and the Affective Neuroscience Personality Scales.
- The affective brain and core-consciousness: How does neural activity generate emotional feelings?
- Affective neuroscience: The foundations of human and animal emotions.
- Affective consciousness: Core emotional feelings in animals and humans.
- A useful animal model for social-affective neuroscience research: Basic description of separation distress, social attachments and play.
- Social neuroscience: Integrating biological and psychological explanations of

social behavior.

- The emotional brain: The mysterious underpinnings of emotional life.

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ИТОГОВЫЙ ВЫВОД ПО КНИГЕ

- Эмоции – не вторичный продукт мышления, а фундаментальная биологическая архитектура психики.

- Понимание базовых аффективных систем помогает точнее видеть природу психических расстройств и источники благополучия.

- Нейроэволюционный подход объединяет биологию, психологию и философию сознания в единую рабочую модель.

КОНЕЦ КОНСПЕКТА